

Lecture 4

Topics: Karnaugh Maps

cd \ ab	00	01	11	10
00	1	0	1	1
01	0	0	1	0
11	0	1	1	0
10	1	1	1	1

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Readings

- David Tarnoff, Computer Organization and Design Fundamentals (CHAPTER 7)

Simplification of circuit

- Goals:
 - minimize the cost of realizing a Boolean function
- How do we measure cost of a circuit?
 - Number of gates
 - Number of levels
 - Gate fan in and/or fan out
 - Interconnection complexity
 - Preventing hazards

How do we simplify?

- Two Primary Methods-
 - **Boolean algebra**: a set of rules that let us transform Boolean expressions into equivalent Boolean expressions.
 - **Karnaugh maps** (also called **K-maps**): similar to truth tables.
- The Karnaugh-map method provides a step-by-step procedure. If you follow the steps correctly, you'll get the right answer. The Boolean-algebra method requires more ingenuity on your part but may be quicker.
- K-maps outputs the circuit in **Minimum** SOP or POS Representations

Karnaugh-map – 2 steps

- Step 1: Construct the Karnaugh-map
- Step 2: Solve equation using Karnaugh-map

Karnaugh-map - graphical representations of truth tables

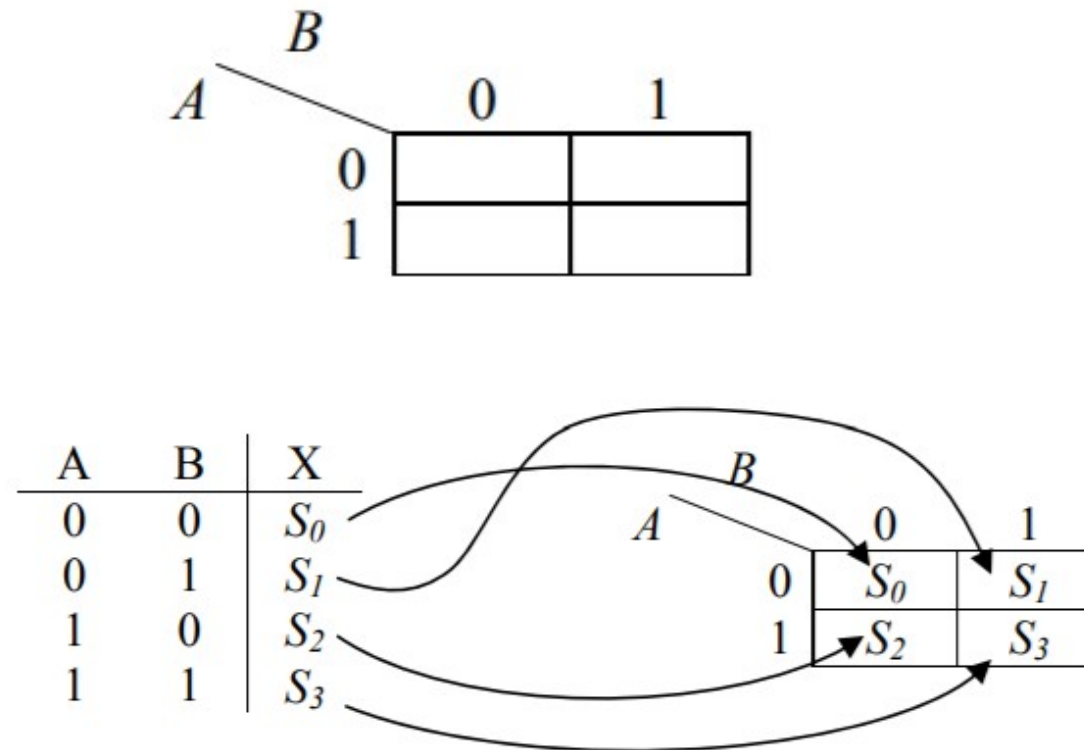


figure 7-2 Mapping a 2-Input Truth Table to Its Karnaugh Map

3-input Karnaugh Map

<i>C</i>		0	1
<i>AB</i>	00		
	01		
	11		
	10		

Figure 7-3 Three-Input Karnaugh Map

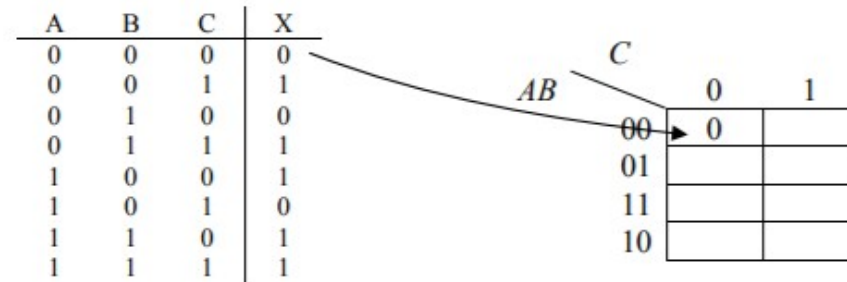
4-input Karnaugh Map

<i>AB</i> \ <i>CD</i>	00	01	11	10
00				
01				
11				
10				

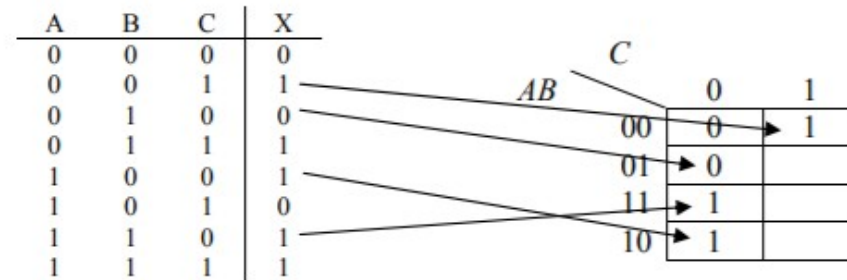
Figure 7-4 Four-Input Karnaugh Map

Convert the three-input truth table below to its corresponding Karnaugh map.

A	B	C	X
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	1
1	0	0	1
1	0	1	0
1	1	0	1
1	1	1	1



A few more of the transfers are shown in the map below.

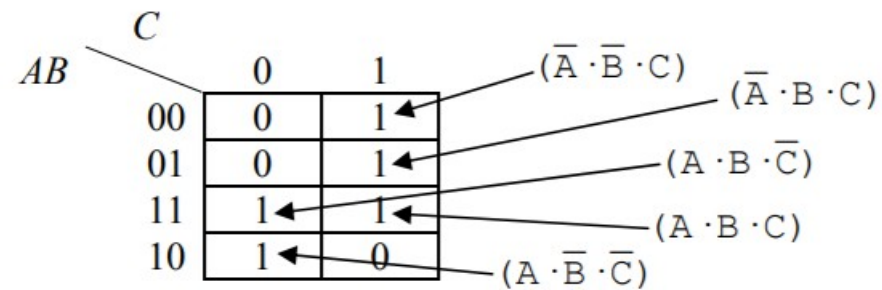


The final map is shown below.

		C	
		0	1
AB	00	0	1
	01	0	1
	11	1	1
	10	1	0

Using Karnaugh Maps

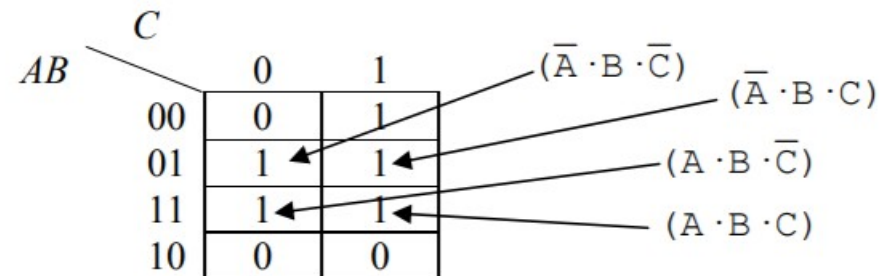
$$X = (\bar{A} \cdot \bar{B} \cdot C) + (\bar{A} \cdot B \cdot C) + (A \cdot B \cdot \bar{C}) + (A \cdot B \cdot C) + (A \cdot \bar{B} \cdot \bar{C})$$



A 2x4 Karnaugh map for variables A, B, and C. The vertical axis is labeled AB with values 00, 01, 11, 10. The horizontal axis is labeled C with values 0, 1. The map contains 1s at (00,1), (01,1), (11,0), (11,1), and (10,0). Five product terms are identified with arrows pointing to the 1s: $(\bar{A} \cdot \bar{B} \cdot C)$ points to (00,1), $(\bar{A} \cdot B \cdot C)$ points to (01,1), $(A \cdot B \cdot \bar{C})$ points to (11,0), $(A \cdot B \cdot C)$ points to (11,1), and $(A \cdot \bar{B} \cdot \bar{C})$ points to (10,0).

		0	1	
AB	00	0	1	$(\bar{A} \cdot \bar{B} \cdot C)$
	01	0	1	$(\bar{A} \cdot B \cdot C)$
	11	1	1	$(A \cdot B \cdot \bar{C})$
	10	1	0	$(A \cdot B \cdot C)$
				$(A \cdot \bar{B} \cdot \bar{C})$

Figure 7-5 Identifying the Products in a Karnaugh Map



A 2x4 Karnaugh map for variables A, B, and C. The vertical axis is labeled AB with values 00, 01, 11, 10. The horizontal axis is labeled C with values 0, 1. The map contains 1s at (00,1), (01,0), (01,1), (11,0), and (11,1). Four product terms are identified with arrows pointing to the 1s: $(\bar{A} \cdot B \cdot \bar{C})$ points to (01,0), $(\bar{A} \cdot B \cdot C)$ points to (01,1), $(A \cdot B \cdot \bar{C})$ points to (11,0), and $(A \cdot B \cdot C)$ points to (11,1).

		0	1	
AB	00	0	1	$(\bar{A} \cdot B \cdot \bar{C})$
	01	1	1	$(\bar{A} \cdot B \cdot C)$
	11	1	1	$(A \cdot B \cdot \bar{C})$
	10	0	0	$(A \cdot B \cdot C)$

Using Karnaugh Maps (cont.)

		C		
		0	1	
AB	00	0	0	$(\bar{A} \cdot B \cdot \bar{C})$
	01	1	1	$(\bar{A} \cdot B \cdot C)$
	11	1	1	$(A \cdot B \cdot \bar{C})$
	10	0	0	$(A \cdot B \cdot C)$

$$X = \bar{A} \cdot B \cdot \bar{C} + \bar{A} \cdot B \cdot C + A \cdot B \cdot \bar{C} + A \cdot B \cdot C$$

$$X = \bar{A} \cdot B \cdot (\bar{C} + C) + A \cdot B \cdot (\bar{C} + C)$$

$$X = \bar{A} \cdot B + A \cdot B$$

$$X = B \cdot (\bar{A} + A)$$

$$X = B$$

Using Karnaugh Maps – forming groups

- The key to effectively using Karnaugh maps is to find the largest group of adjacent cells containing ones
 - The larger the group, the fewer products and inputs will be needed to create the boolean expression that produces the truth table

Using Karnaugh Maps – forming groups (Cont.)

- The grouping must be in the shape of a rectangle. There are no diagonal adjacencies allowed.

		<i>CD</i>			
		00	01	11	10
<i>AB</i>	00	0	0	0	1
	01	1	1	0	1
	11	1	1	0	1
	10	0	0	0	1

Right

		<i>CD</i>			
		00	01	11	10
<i>AB</i>	00	0	0	0	0
	01	0	0	1	0
	11	0	1	0	0
	10	0	0	0	0

Wrong

Using Karnaugh Maps – forming groups (Cont.)

- All cells in a rectangle must contain ones. No zeros are allowed.

		<i>CD</i>			
		00	01	11	10
<i>AB</i>	00	1	0	0	0
	01	1	0	1	1
	11	1	0	1	1
	10	1	0	0	0

Right

		<i>CD</i>			
		00	01	11	10
<i>AB</i>	00	0	0	0	0
	01	0	1	1	0
	11	0	1	0	0
	10	0	0	0	0

Wrong

Using Karnaugh Maps – forming groups (Cont.)

- The number of cells in the grouping must equal a power of two, i.e., only groups of 1, 2, 4, 8, or 16 are allowed.

		<i>CD</i>			
		00	01	11	10
<i>AB</i>	00	1	0	1	1
	01	0	0	1	1
	11	0	1	1	1
	10	0	0	1	1

Right

		<i>CD</i>			
		00	01	11	10
<i>AB</i>	00	1	1	0	0
	01	1	1	0	0
	11	1	1	0	0
	10	0	0	0	0

Wrong

Using Karnaugh Maps – forming groups (Cont.)

- Outside edges of Karnaugh maps are considered adjacent, so rectangles may wrap from left to right or from top to bottom.

		<i>CD</i>			
		00	01	11	10
<i>AB</i>	00	0	1	1	0
	01	0	0	0	0
	11	1	0	0	1
	10	0	1	1	0

Using Karnaugh Maps – forming groups (Cont.)

- Cells may be contained in more than one rectangle, but every rectangle must have at least one cell unique to it. (In wrong example, the horizontal rectangle is an unnecessary duplicate.)

		<i>CD</i>			
<i>AB</i>		00	01	11	10
	00	0	1	1	0
	01	1	1	1	1
	11	1	1	1	1
	10	0	0	0	1

Right

		<i>CD</i>			
<i>AB</i>		00	01	11	10
	00	0	0	0	0
	01	0	1	0	0
	11	0	1	1	0
	10	0	0	1	0

Wrong

Using Karnaugh Maps – forming groups (Cont.)

- Every rectangle must be as large as possible.

		<i>CD</i>			
		00	01	11	10
<i>AB</i>	00	1	1	0	0
	01	1	1	0	0
	11	1	1	0	0
	10	1	1	0	0

Right

		<i>CD</i>			
		00	01	11	10
<i>AB</i>	00	1	1	0	0
	01	1	1	0	0
	11	1	1	0	0
	10	1	1	0	0

Wrong

Using Karnaugh Maps – forming groups (Cont.)

- Every 1 must be covered by at least one rectangle.

		<i>CD</i>			
		00	01	11	10
<i>AB</i>	00	1	0	0	1
	01	1	1	1	1
	11	1	1	1	1
	10	1	0	0	0

Right

		<i>CD</i>			
		00	01	11	10
<i>AB</i>	00	1	0	0	1
	01	1	1	1	1
	11	1	1	1	1
	10	1	0	0	0

Wrong

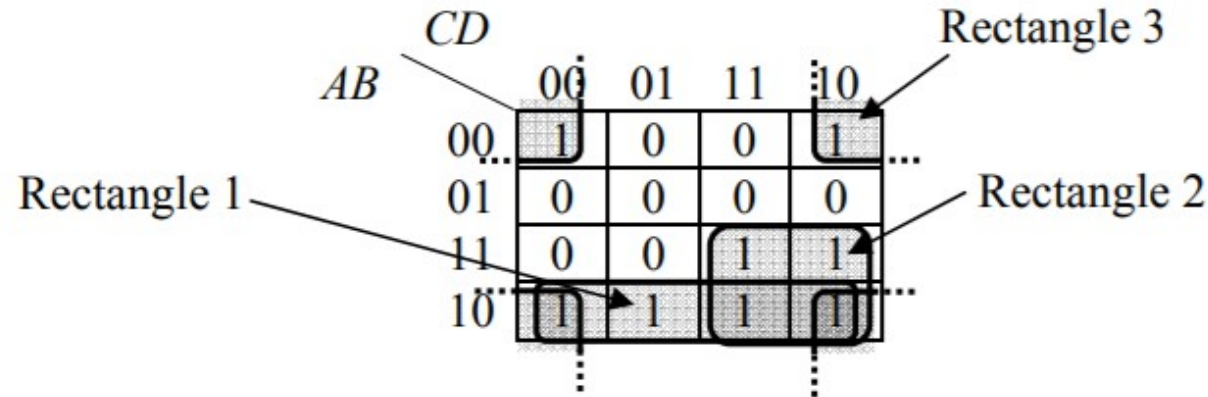
Determine the minimal SOP expression for the truth table below

A	B	C	D	X
0	0	0	0	1
0	0	0	1	0
0	0	1	0	1
0	0	1	1	0
0	1	0	0	0
0	1	0	1	0
0	1	1	0	0
0	1	1	1	0
1	0	0	0	1
1	0	0	1	1
1	0	1	0	1
1	0	1	1	1
1	1	0	0	0
1	1	0	1	0
1	1	1	0	1
1	1	1	1	1

First, we need to convert the truth table to a Karnaugh map.

		<i>CD</i>			
		00	01	11	10
<i>AB</i>	00	1	0	0	1
	01	0	0	0	0
	11	0	0	1	1
	10	1	1	1	1

Determine the minimal SOP expression for the truth table below (Cont.)



Rectangle 1:

A	B	C	D
1	0	0	0
1	0	0	1
1	0	1	1
1	0	1	0

Product for rectangle 1 = $A \cdot \overline{B}$

Rectangle 2:

A	B	C	D
1	1	1	1
1	1	1	0
1	0	1	1
1	0	1	0

Product for rectangle 2 = $A \cdot C$

Rectangle 3:

A	B	C	D
0	0	0	0
0	0	1	0
1	0	0	0
1	0	1	0

Product for rectangle 3 = $\overline{B} \cdot \overline{D}$

$$X = A \cdot \overline{B} + A \cdot C + \overline{B} \cdot \overline{D}$$

Determine the minimal SOP expression for the truth table below

A	B	C	X
0	0	0	0
0	0	1	1
0	1	0	1
0	1	1	0
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	0

A	B	C	X
0	0	0	1
0	0	1	1
0	1	0	1
0	1	1	0
1	0	0	0
1	0	1	0
1	1	0	1
1	1	1	1

A	B	C	X
0	0	0	1
0	0	1	0
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	1

"Don't Care" Conditions in a Karnaugh Map

- Don't cares in a Karnaugh map, or truth table, may be either 1s or 0s, as long as **we don't care what the output is for an input condition we never expect to see**. We plot these cells with an x, among the normal **1s** and **0s**.
- Some combinations of input signals are completely inconsequential to the proper operation of the circuit. It is possible to take advantage of these situations to further minimize logic circuits.

"Don't Care" Conditions in a Karnaugh Map (Cont.)

		<i>CD</i>			
		00	01	11	10
<i>AB</i>	00	1	0	0	0
	01	1	0	0	X
	11	1	0	1	1
	10	X	0	0	0

Figure 7-8 Karnaugh Map with a "Don't Care" Elements

		<i>CD</i>			
		00	01	11	10
<i>AB</i>	00	1	0	0	0
	01	1	0	0	X
	11	1	0	1	1
	10	X	0	0	0

Figure 7-9 Karnaugh Map with a "Don't Care" Elements Assigned

Determine the minimal SOP expression for the K-Map below

		<i>CD</i>			
		00	01	11	10
<i>AB</i>	00	0	0	0	0
	01	1	1	0	X
	11	1	X	1	1
	10	X	0	0	0

Combinational Logic Applications

- Full adder
- Seven-Segment Displays
- Decoders
- Multiplexers
- Demultiplexers